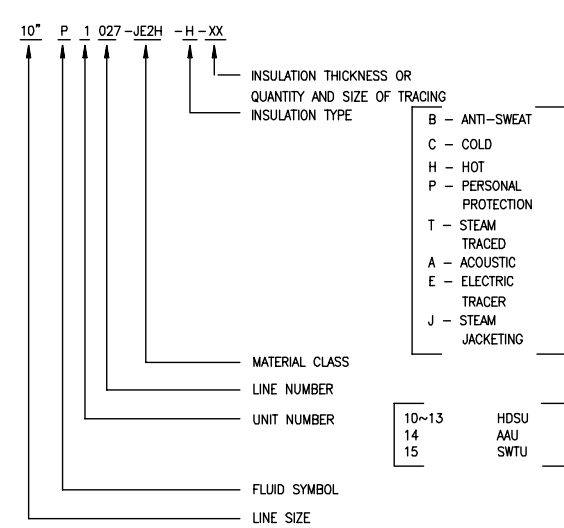


PIPING SYMBOLS AND LEGENDS

	MAIN PROCESS LINES
	OTHER PROCESS AND UTILITY LINES
	PROCESS LINES WITH STEAM TRACING
	WATER SPARY
	PROCESS LINES WITH STEAM JACKETED
	GATE VALVE
	BALL VALVE
	DIAPHRAGM VALVE
	BUTTERFLY VALVE
	GLOBE VALVE
	CHECK VALVE
	STOP CHECK STRAIGHT VALVE
	STOP CHECK ANGLE VALVE
	PLUG VALVE
	THREE WAY PLUG VALVE
	NEEDLE VALVE
	JACKET VALVE
	SLIDE VALVE
	AUTOMATIC RECIRCULATION VALVE (ARV)
	ORBIT VALVE
	SPRING LOADED QUICK CLOSE VALVE FOR INTERMITTENT BLOWDOWN
	CONTINUOUS BLOWDOWN
	GATE VALVE WITH GLOBE BYPASS VALVE
	LINE CLASS CHANGE
	ANGLE VALVE
	THREE-WAY VALVE
	FOUR-WAY VALVE
	EQUALIZING VALVE ON MAIN VALVE BODY
	WELD CAP
	BLIND FLANGE
	HOSE CONNECTION
	REMOVABLE PIPE SPOOL
	SCREWED CAP
	FLANGE JOINT
	BURNER NOZZLE
	REDUCTION IN LINE SIZE
	PULSATION DAMPENR
	FREE DRAIN LINE (DO NOT POCKET LINE)
	SLOPED LINE
	FLEXIBLE HOSE
	EXPANSION JOINT

	CONTINUITY ARROW
	CONTINUITY ARROW (FROM BATTERY LIMIT)
	SPACER
	SPACER (NORMALLY CLOSED)
	SPECTACLE BLIND (NORMALLY OPEN)
	SPECTACLE BLIND (NORMALLY CLOSED)
	DAMPNER
	OIL SEPARATOR
	SIGHT GLASS
	DRESSER COUPLING
	STEAM TRAP
	FILTER
	STRAINER (B - BASKET, C - CONE, P - PLATE, T - TEMPORARY)
	BASKET TYPE STRAINER
	TEE TYPE STRAINER
	"Y" TYPE STRAINER
	STRAINER-DUAL
	VORTEX BREAKER
	MANOMETER
	VENT
	OPEN VENT
	LIQUID SEAL
	BREATHER VALVE
	FLAME ARRESTER
	PIPING SPECIALTY ITEM
	SAMPLE CONNECTION WITHOUT COOLER
	SAMPLE CONNECTION WITH COOLER
	TRAP-VACUUM BOOSTER (LIFT)
	CONTINUOUS DRAINER
	STEAM EXHAUST HEAD
	GAS TIGHT GAUGE HATCH (WITHOUT VENTING GAS TO ATMOSPHERE)
	GAUGE HATCH
	DRAIN CONN. TO CLOSED SYSTEM
	AREA DRAIN FITTING
	OILY DRAIN
	EQUIPMENT INSULATION
	PACKAGE EQUIPMENT OR INTEGRAL UNITS
	SUPPLY LIMIT
	LINE CLASS BREAK
	INSULATION BREAK
	LINE NUMBER BREAK
	SPRAY NOZZLE DEVICE

LINE NOMENCLATURE



LINE CLASS DESIGNATION

1ST SYMBOL: RATING AND FACING

A:	ANSI 150 LB, RF
B:	ANSI 300 LB, RF
C:	ANSI 600 LB, RF
D:	ANSI 900 LB, RF
E:	ANSI 1500 LB, RF
F:	ANSI 2500 LB, RF
G:	ANSI 2500 LB, RJ
H:	ANSI 1500 LB, RJ
I:	ANSI 900 LB, RJ
J:	ANSI 600 LB, RJ
K:	ANSI 300 LB, RJ
L:	ANSI 150 LB, /125 LB, FF

2ND SYMBOL: PIPE MATERIAL

A:	CARBON STEEL
B:	KILLED CARBON STEEL
C:	CAST IRON
D:	CARBON STEEL, GALVANIZED
E:	1 1/4 CR-1/2 MO STEEL
F:	2 1/4 CR-1 MO STEEL
G:	5 CR-1/2 MO STEEL
H:	9 CR-1 MO STEEL
J:	304 STAINLESS STEEL
K:	316 STAINLESS STEEL
L:	304L STAINLESS STEEL
M:	316L STAINLESS STEEL
N:	321 STAINLESS STEEL
P:	347 STAINLESS STEEL
R:	CARBON STEEL (LOW TEMP. SERVICE)
S:	3 1/2 NI STEEL
T:	C.S. WITH PTFE LINING
U:	304H STAINLESS STEEL

3RD SYMBOL: CORROSION ALLOWANCE

0:	0" (0 MM)
1:	0.039" (1.0 MM)
2:	0.063" (1.6 MM)
3:	0.100" (2.5 MM)
4:	0.125" (3.0 MM)
5:	0.188" (4.8 MM)
6:	0.250" (6.3 MM)

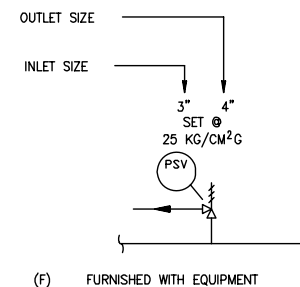
4TH SYMBOL: SPECIAL SERVICE

A:	AMINE
B:	ANSI B.31.1 PIPING
C:	HYDROGEN AND SOUR SERVICE
H:	HYDROGEN
J:	JACKETING
S:	NACE SOUR SERVICE
U:	UTILITY (STEAM, AIR,NITROGEN, AND CONDENSATE ETC.)
W:	FIREWATER, INDUSTRIAL WATER, COOLING WATER
X:	JACKETING AND SOUR SERVICE

VALVE POSITION

CSC	CAR SEAL CLOSED
CSO	CAR SEAL OPEN
FC	FAIL CLOSED
FL	FAILURE LOCK
FLC	FAILURE LOCK TO CLOSE
FO	FAIL OPEN
LC	LOCKED CLOSED
LO	LOCKED OPEN
NC	NORMALLY CLOSED (OR)
NO	NORMALLY OPEN
TSO	TIGHT SHUT-OFF(NOTE 5)

LEGEND FOR SAFETY VALVE



FLUID SYMBOL FOR LINE SERVICE

SERVICE SYSTEM	FLUID	SYMBOL
AIR SYSTEM	INSTRUMENT AIR	IA
	PLANT AIR	PA
	DRY AIR	DA
BLOWDOWN AND PUMPOUT	INTERMITTENT BLOWDOWN	BB
	CONTINUOUS BLOWDOWN	CB
	SLOP	SB
	BLOWDOWN	BD
DRAIN	CHEMICAL OR CONTAMINATED DRAIN	CD
	AMINE DRAIN	ED
	LIQUID DRAIN	LD
	OIL DRAIN	OD
	OILY SEWER	OS
	NON-CONTAMINATED DRAIN	ND
	SOUR WATER DRAIN	WD
	ACID DRAIN	AD
	WATER SEWER	WS
	SPENT CAUSTIC DRAIN	CLD
FLARE SYSTEM	H ₂ S FLARE	XF
	HOT FLARE	HF
	WET FLARE	WF
	DRY FLARE	DF
GAS SYSTEM	FUEL GAS	FG
	NITROGEN	NG
	NATURAL GAS	GG
OIL UTILITY SYSTEM	FLUSH OIL	FO
	SEAL OIL	SO
	FLUSHING OIL PURGE OIL	WO
	HYDRAULIC OIL	HO
	INJECTION OIL	IO
	LUBE OIL	LO
	QUENCH OIL	QO
	FLUX OIL	XO
	DISPLACEMENT NAPHTHA (ATMOSPHERIC NAPHTHA)	DN
	DISPLACEMENT OIL (ATMOSPHERIC GAS OIL)	DO
PROCESS	PROCESS SERVICE	P
STEAM AND CONDENSATE	HIGH PRESSURE CONDENSATE	HC
	MEDIUM PRESSURE CONDENSATE	MC
	LOW PRESSURE CONDENSATE	LC
	HIGH PRESSURE STEAM	HS
	HIGH PRESSURE SUPERHEATED STEAM	XS
	MEDIUM PRESSURE STEAM	MS
	MEDIUM PRESSURE SUPERHEATED STEAM	TS
	LOW PRESSURE STEAM	LS
	LOW PRESSURE SUPERHEATED STEAM	SS
	EXHAUSTED STEAM	ES
	DILUTION STEAM	DS
VENT	ATMOSPHERE	AV
	SAFETY VALVE TO ATMOSPHERE	SV
WATER SYSTEM	COOLING WATER SUPPLY	SW
	COOLING WATER RETURN	RW
	TEMPERED WATER SUPPLY	LW
	INTERMEDIATE COOLING WATER	MW
	TEMPERED WATER RETURN	LW
	FIRE WATER	FW
	DRINK WATER	DW
	UTILITY WATER (INDUSTRIAL WATER)	UW
	TREATED WATER (DEMINERALIZED WATER)	TW
	DEAERATED BOILER FEED WATER	BW
	WASH WATER (FLUSH WATER & PURGE WATER)	XW
	PROCESS WATER	PW
	WASTE WATER	WW
	QUENCH WATER	QW
SPECIAL LIQUIDS	CAUSTIC	CL
	CHEMICAL INJECTION	IL
	ACID	AL
	MEA, DIPA, DEA	EL
	SOLVENT	SL
	SULFOLANE	SU
	PDEB	DE

GENERAL NOTES:

1. THE DRAIN VALVE UPSTREAM THE CONTROL VALVE SHALL BE ¾"
2. PUMP CASING VENT AND DRAIN PIPE SHALL BE EXTENDED TO OILY SEWER.
3. HIGH POINTS OF ALL LINES SHALL BE VENTED AND LOW POINTS DRAINED EXCEPT WHERE THE LINE CAN BE VENTED OR DRAINED ELSE WHERE AND GENERALLY, ABOVE VENTS END DRAINS ARE NOT NOTED ON THE P & I DIAGRAM.
4. A BLEED VALVE FOR A PROCESS BLOCK VALVE SHALL BE LOCATED AS CLOSE AS POSSIBLE TO THE BLOCK VALVE.
5. THE LEAKAGE RATE OF TIGHTLY SHUT-OFF CONTROL VALVE SHALL BE BETTER THAN ANSI CLASS V.

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